

Identifying Customer Needs



Courtesy of Nest Labs

EXHIBIT 5-1

The Nest learning thermostat, which programs itself based on patterns of activity in the home.

Tony Fadell had been a product development leader at Apple for more than a decade, participating in the creation of the iPod portable music player and the iPhone smartphone. After leaving Apple, he contemplated entrepreneurial opportunities and eventually decided to create a better thermostat for control of home heating and cooling systems. Exhibit 5-1 shows the Nest Learning Thermostat, the first product he and his team created from their new company, Nest Labs.

The Nest team aspired to create the best thermostat on the market, and knew that understanding and delivering on customer needs was key to success. The Nest thermostat was extremely well received in the marketplace, and just three years after its formation, Nest Labs was acquired by Google for \$3.2 billion, representing a phenomenal financial outcome for the young company.

This chapter presents a method for comprehensively identifying a set of customer needs, and is built around the example of creating a better thermostat.

The goals of the method are to:

- Ensure that the product is focused on customer needs.
- Identify latent or hidden needs as well as explicit needs.
- Provide a fact base for justifying the product specifications.
- Create an archival record of the needs activity of the development process.
- Ensure that no critical customer need is missed or forgotten.
- Develop a common understanding of customer needs among members of the development team.

The philosophy behind the method is to create a high-quality information channel that runs directly between customers in the target market and the developers of the product. This philosophy is built on the premise that those who directly control the details of the product, including the engineers and industrial designers, must interact with customers and experience the *use environment* of the product. Without this direct experience, technical trade-offs are not likely to be made correctly, innovative solutions to customer needs may never be discovered, and the development team may never develop a deep commitment to meeting customer needs.

The process of identifying customer needs is an integral part of the larger product development process and is very closely related to opportunity identification, product planning, concept generation, concept selection, competitive benchmarking, and the establishment of product specifications. The customer-needs activity is shown in Exhibit 5-2 in

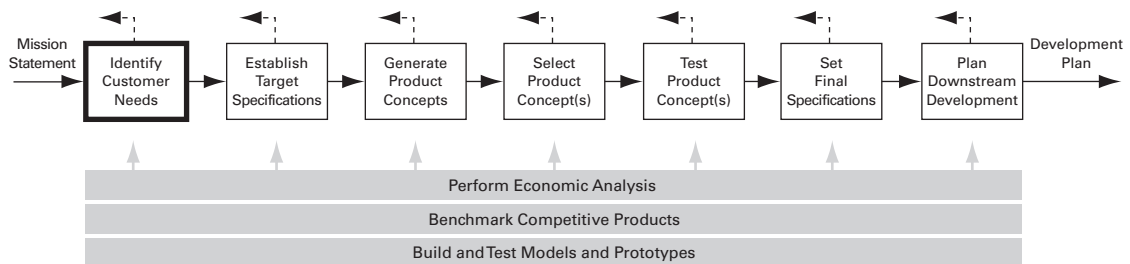


EXHIBIT 5-2 The customer-needs activity in relation to other concept development activities.

relation to these other front-end product development activities, which collectively can be thought of as the *concept development* phase.

The concept development process illustrated in Exhibit 5-2 implies a distinction between customer needs and product specifications. This distinction is subtle but important. *Needs* are largely independent of any particular product we might develop; they are not specific to the concept we eventually choose to pursue. A team should be able to identify customer needs without knowing if or how it will eventually address those needs. On the other hand, *specifications* do depend on the concept we select. The specifications for the product we finally choose to develop will depend on what is technically and economically feasible and on what our competitors offer in the marketplace, as well as on customer needs. (See Chapter 6, Product Specifications, for a more detailed discussion of this distinction.) Also note that we choose to use the word *need* to label any attribute of a potential product that is desired by the customer; we do not distinguish here between a want and a need. Other terms used in industrial practice to refer to customer needs include *customer attributes* and *customer requirements*.

The Importance of Latent Needs

Latent needs are those not yet widely recognized by most customers, and not yet fulfilled by existing products. The needs exist, and if fulfilled, would result in greater customer satisfaction, yet they remain largely unknown. For instance, prior to the year 2000, mobile telephones did not contain cameras. Most consumers did not know that they wanted to be able to take photographs with their telephones; however, that need was real, and became a differentiating feature of telephones that fulfilled it. The ability to recognize latent needs is therefore a critical capability in product development, giving firms the ability to create products that surprise and delight customers. Of course, once latent needs are revealed through successful products that fulfill them, they become widely adopted and become “must haves.” By 2010, essentially all mobile computing devices could take photographs.

The Process of Identifying Customer Needs

Identifying customer needs is itself a process, for which we present a five-step method. We believe that a little structure goes a long way in facilitating effective product development practices, and we hope and expect that this method will be viewed by those who employ it not as a rigid process but rather as a starting point for continuous improvement and refinement. The five steps are:

1. Gather raw data from customers.
2. Interpret the raw data in terms of customer needs.
3. Organize the needs into a hierarchy of primary, secondary, and (if necessary) tertiary needs.
4. Establish the relative importance of the needs.
5. Reflect on the results and the process.

In this chapter, we treat each of the five steps in turn and illustrate key points with the thermostat example. We chose the thermostat because it is simple enough that the method is not

hidden by the complexity of the example; however, note that the same method, with minor adaptation, has been successfully applied to many kinds of products ranging from kitchen utensils costing less than \$10 to machine tools costing hundreds of thousands of dollars.

Before beginning the development project, the firm typically specifies a particular market opportunity and lays out the broad constraints and objectives for the project. This information is frequently formalized as a *mission statement* (also sometimes called a *charter* or a *design brief*). The mission statement specifies which direction to go in but generally does not specify a precise destination or a particular way to proceed. The mission statement is the result of the product planning activities described in Chapter 4, Product Planning. The mission statement for a thermostat is shown in Exhibit 5-3.

The thermostat category of products is already relatively well developed. Such products are particularly well suited to a structured process for gathering customer needs. One could reasonably ask whether a structured method is effective for completely new categories of products with which customers have no experience. Satisfying needs is just as important in revolutionary products as in incremental products. A necessary condition for product success is that a product offer perceived benefits to the customer. Products offer benefits when they satisfy needs. This is true whether the product is an incremental variation on an existing product or whether it is a completely new product based on a revolutionary invention. Developing an entirely new category of product is a risky undertaking, and to some extent the only real indication of whether customer needs have been identified correctly is whether customers like the team's first prototypes. Nevertheless, in our opinion, a structured method for gathering data from customers remains useful and can lower the inherent risk in developing a radically new product. Whether or not customers are able to fully articulate their true needs, interaction with customers in the target market will help the development team build a personal understanding of the user's environment and point of view. This information is always useful, even if it does not result in the identification of every need the new product will address.

EXHIBIT 5-3

Mission statement for the thermostat.

Mission Statement: Thermostat Project	
Product Description	• Thermostat for residential use.
Benefit Proposition	• Simple to use, attractive, and saves energy.
Key Business Goals	• Product introduced in fourth quarter of 2012. • 50% gross margin. • 10% share of replacement thermostat market by 2016.
Primary Market	• Residential consumer.
Secondary Markets	• Heating, ventilation, and air conditioning contractors doing residential work.
Assumptions	• Replacement for an existing thermostat. • Compatible with most existing systems and wiring.
Stakeholders	• User • Retailer • Sales force • Service center • Production • Legal department

Step 1: Gather Raw Data from Customers

Consistent with our basic philosophy of creating a high-quality information channel directly from the customer, gathering data involves contact with customers and experience with the use environment of the product. Three methods are commonly used:

1. Interviews: One or more development team members discusses needs with a single customer. Interviews are usually conducted in the customer's environment and typically last one to two hours.

2. Focus groups: A moderator facilitates a two-hour discussion with a group of 8 to 12 customers. Focus groups are typically conducted in a special room equipped with a transparent mirror allowing several members of the development team to observe the group. In most cases, the moderator is a professional market researcher, but a member of the development team sometimes moderates. The proceedings are usually video recorded. Participants are usually paid a modest fee (\$50 to \$100 each) for their attendance. The total cost of a focus group, including rental of the room, participant fees, video recording, and refreshments, is about \$5,000. In most U.S. cities, firms that recruit participants, moderate focus groups, and/or rent facilities are listed in directories and search engines under "Market Research."

3. Observing the product in use: Watching customers use an existing product or perform a task for which a new product is intended can reveal important details about customer needs. For example, watching a consumer replace an existing thermostat may reveal a mismatch in shapes between the new and old products causing unsightly discontinuities in the wall surface. Observation may be completely passive, without any direct interaction with the customer, or may involve working side by side with a customer, allowing members of the development team to develop firsthand experience using the product. Ideally, team members observe the product in the actual use environment. Procter & Gamble, for example, observes thousands of customers every year in their homes or workplaces to better understand their needs. For some products, such as do-it-yourself tools, actually using the products is simple and natural; for others, such as surgical instruments, the team may have to use the products on surrogate tasks (e.g., cutting fruit instead of human tissue when developing a new scalpel).

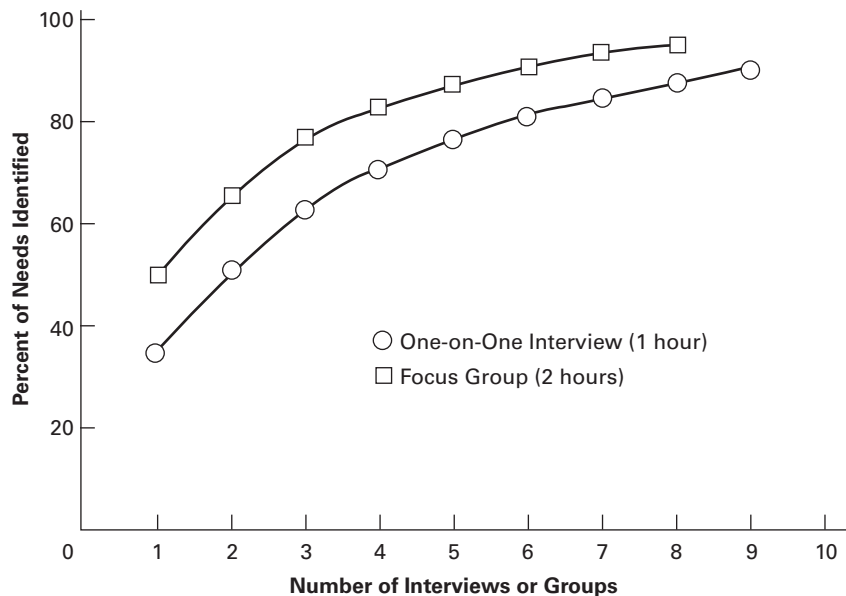
Some practitioners also rely on written surveys for gathering raw data. While a Web-based survey is quite useful later in the process, we cannot recommend this approach for initial efforts to identify customer needs; text-based surveys simply do not provide enough information about the use environment of the product, and they are generally ineffective in revealing unanticipated needs.

Research by Griffin and Hauser shows that one 2-hour focus group reveals about the same number of needs as two 1-hour interviews (Griffin and Hauser, 1993). (See Exhibit 5-4.) Because interviews are usually less costly (per hour) than focus groups and because an interview often allows the product development team to experience the use environment of the product, we recommend that interviews be the primary data collection method. Interviews may be supplemented with one or two focus groups as a way to allow top management to observe a group of customers or as a mechanism for sharing a common customer experience (via video) with the members of a larger team. Some practitioners believe that for certain products and customer groups, the interactions among the participants of focus groups can elicit more varied needs than are revealed through interviews, although this belief is not strongly supported by research findings.

EXHIBIT 5-4

Comparison of the percentages of customer needs that are revealed for focus groups and interviews as a function of the number of sessions. Note that a focus group lasts two hours, while an interview lasts one hour.

Source: Abbie Griffin and John R. Hauser, "The Voice of the Customer," *Marketing Science*, Vol. 12, No. 1, Winter 1993, pp. 1–27



Choosing Customers

Griffin and Hauser also addressed the question of how many customers to interview to reveal most of the customer needs. In one study, they estimated that 90 percent of the customer needs for picnic coolers were revealed after 30 interviews. In another study, they estimated that 98 percent of the customer needs for a piece of office equipment were revealed after 25 hours of data collection in both focus groups and interviews. As a practical guideline for most products, conducting fewer than 10 interviews is probably inadequate and 50 interviews are probably too many; however, interviews can be conducted sequentially and the process can be terminated when no new needs are revealed by additional interviews. These guidelines apply to cases in which the development team is addressing a single market segment. If the team wishes to gather customer needs from multiple distinct segments, then the team may need to conduct 10 or more interviews in each segment. Concept development teams consisting of more than 10 people usually collect data from plenty of customers simply by involving much of the team in the process. For example, if a 10-person team is divided into five pairs and each pair conducts 6 interviews, the team conducts 30 interviews in total.

Needs can be identified more efficiently by interviewing *lead users* and/or extreme users. According to von Hippel, lead users are customers who experience needs months or years ahead of the majority of the market and stand to benefit substantially from product innovations (von Hippel, 1988). These customers are particularly useful sources of data for two reasons: (1) they are often able to articulate their emerging needs, because they have had to struggle with the inadequacies of existing products, and (2) they may have already invented solutions to meet their needs. By focusing a portion of the data collection efforts on lead users, the team may be able to identify needs that, although explicit for lead users, are still latent for the majority of the market. Developing products to meet these latent needs allows a firm to anticipate trends and to leapfrog competitive products.

EXHIBIT 5-5

Customer
selection matrix
for a thermostat
project.

	Lead Users	Users	Retailers
Homeowner	5	5	5
HVAC Contractor	5	5	

Gary Burchill, et al., Concept Engineering, Center for Quality of Management, Cambridge, MA, Document No. ML0080, 1997

Extreme users are those who use the product in unusual ways or who have special needs. For example, extreme users of the thermostat might be people who have limited vision or dexterity or those who need to vary temperatures many times each day. Extreme users can help the team identify needs that may be felt less acutely by the mainstream market, but are nevertheless important opportunities for competitive advantage. For example, entrepreneur Sam Farber created the original Good Grips vegetable peeler in response to the needs of his wife who suffered from arthritis. Her extreme needs proved to be a reflection of the latent need for more ergonomic kitchen tools among mainstream users.

The choice of which customers to interview is complicated when several different groups of people can be considered “the customer.” For many products, one person (the buyer) makes the buying decision and another person (the user) actually uses the product. A good approach is to gather data from the end user of the product in all situations, and in cases where other types of customers and stakeholders are clearly important, to gather data from these people as well.

A customer selection matrix is useful for planning exploration of both market and customer variety. Burchill suggests that market segments be listed on the left side of the matrix while the different types of customers are listed across the top (Burchill et al., 1997), as shown in Exhibit 5-5. The number of intended customer contacts is entered in each cell to indicate the depth of coverage.

For industrial and commercial products, actually locating customers is usually a matter of making telephone calls or sending e-mail. In developing such products within an existing firm, a field sales force can often provide names of customers, although the team must be careful about biasing the selection of customers toward those with allegiances to a particular manufacturer. Online directories can be used to identify names of some types of customers for some classes of products (e.g., building contractors or insurance agents). For products that are integral to a customer’s job, getting someone to agree to an interview is usually simple; these customers are eager to discuss their needs. For consumer products, customers can also be located through groups on social networks and via e-mail; however, arranging a set of interviews for consumer products generally requires more inquiries than for industrial or commercial products because the benefit of participating in an interview is less direct for these customers.

The Art of Eliciting Customer Needs Data

The techniques we present here are aimed primarily at interviewing end users, but these methods do apply to all of the three data-gathering modes and to all types of stakeholders. The basic approach is to be receptive to information provided by customers and to avoid confrontations or defensive posturing. Gathering needs data is very different from a sales call: the goal is to elicit an honest expression of needs, not to convince a customer of what he or she needs. In most cases customer interactions will be verbal; interviewers ask questions and the customer responds. A prepared interview guide is valuable for structuring this

dialogue. Some helpful questions and prompts for use after the interviewers introduce themselves and explain the purpose of the interview are:

- When and why do you use this type of product?
- Walk us through a typical session using the product.
- What do you like about the existing products?
- What do you dislike about the existing products?
- What issues do you consider when purchasing the product?
- What improvements would you make to the product?

Here are some general hints for effective interaction with customers:

- ***Go with the flow.*** If the customer is providing useful information, do not worry about conforming to the interview guide. The goal is to gather important data on customer needs, not to complete the interview guide in the allotted time.
- ***Use visual stimuli and props.*** Bring a collection of existing and competitors' products, or even products that are tangentially related to the product under development. At the end of a session, the interviewers might even show some preliminary product concepts to get customers' early reactions to various approaches.
- ***Suppress preconceived hypotheses about the product technology.*** Frequently, customers will make assumptions about the product concept they expect would meet their needs. In these situations, the interviewers should avoid biasing the discussion with assumptions about how the product will eventually be designed or produced. When customers mention specific technologies or product features, the interviewer should probe for the underlying need the customer believes the suggested solution would satisfy.
- ***Have the customer demonstrate the product and/or typical tasks related to the product.*** If the interview is conducted in the use environment, a demonstration is usually convenient and invariably reveals new information.
- ***Be alert for surprises and the expression of latent needs.*** If a customer mentions something surprising, pursue the lead with follow-up questions. Frequently, an unexpected line of questioning will reveal the latent needs, the important dimensions of the customers' needs that are neither fulfilled nor commonly articulated and understood.
- ***Watch for nonverbal information.*** The process described in the chapter is aimed at developing better physical products. Unfortunately, words are not always the best way to communicate needs related to the physical world. This is particularly true of needs involving the human dimensions of the product, such as comfort, image, or style. The development team must be constantly aware of the nonverbal messages provided by customers. What are their facial expressions? How do they hold competitors' products?

Note that many of our suggested questions and guidelines assume that the customer has some familiarity with products similar to the new product under development. This is almost always true. For example, even before the first programmable thermostats became available, people controlled temperature over the course of the day. Developing an understanding of customer needs as they relate to the general temperature control task would still have been beneficial in developing the first programmable device. Similarly, understanding the needs of customers using other types of household devices such as coffee makers or microwave ovens can be informative. We can think of no product so

revolutionary that there would be no analogous products or tasks from which the development team could learn; however, in gathering needs relating to truly revolutionary products with which customers have no experience, the interview questions should be focused on the task or situation in which the new product will be applied, rather than on the product itself.

Documenting Interactions with Customers

Four methods are commonly used for documenting interactions with customers:

1. Audio recording: Making an audio recording of the interview is very easy. Unfortunately, transcribing the recording into text is very time consuming, and hiring someone to do it can be expensive. Also, audio recording has the disadvantage of being intimidating to some customers.

2. Notes: Handwritten notes are the most common method of documenting an interview. Designating one person as the primary note taker allows the other person to concentrate on effective questioning. The note taker should strive to capture some of the wording of every customer statement verbatim. These notes, if transcribed immediately after the interview, can be used to create a description of the interview that is very close to an actual transcript. This debriefing immediately after the interview also facilitates sharing of insights between the interviewers.

3. Video recording: Video recording is almost always used to document a focus group session. It is also very useful for documenting observations of the customer in the use environment and/or using existing products. The video recording is useful for bringing new team members “up to speed” and is also useful as raw material for presentations to upper management. Multiple viewings of video recordings of customers in action often facilitate the identification of latent customer needs. Video recording is also useful for capturing many aspects of the end user’s environment.

4. Still photography: Taking photographs provides many of the benefits of video recording, but is usually less intrusive and therefore easier to do while observing customers in the field. Additional advantages of still photography are ease of display of the photos, excellent image quality, and readily available equipment. The primary disadvantage is the relative inability to record dynamic information.

The final result of the data-gathering phase of the process is a set of raw data, usually in the form of *customer statements* but frequently supplemented by video recordings or photographs. A data template implemented in a spreadsheet is useful for organizing these raw data. Exhibit 5-6 shows an example of a portion of such a template. We recommend that the template be filled in as soon as possible after the interaction with the customer and edited by the other development team members present during the interaction. The first column in the main body of the template indicates the question or prompt that elicited the customer data. The second column is a list of verbatim statements the customer made or an observation of a customer action (from a video recording or from direct observation). The third column contains the customer needs implied by the raw data. Some emphasis should be placed on investigating clues that may identify potential latent needs. Such clues may be in the form of humorous remarks, less serious suggestions, frustrations, nonverbal information, or observations and descriptions of the use environment. The symbol (!) is used in Exhibit 5-6 to flag potential latent needs. Techniques for interpreting the raw data in terms of customer needs are given in the next section.

Customer:	Bill Esposito	Interviewer(s):	Jonathan and Lisa
Address:	100 Memorial Drive Cambridge, MA 02139	Date:	19 January 2015
Telephone:	617-864-1274	Currently uses:	Honeywell Model A45
Email:	bespo@zmail.com	Type of user:	Homeowner
Willing to do follow-up?	Yes		

Question/Prompt	Customer Statement	Interpreted Need
Typical uses	I have to manually turn it on and off when it gets too hot or cold.	The thermostat maintains a comfortable temperature without requiring user action.
	Each time I want to change the temperature, I need to adjust both thermostats in the house.	Any user inputs need not be made in multiple locations. (!)
Likes—current model	I like that I can change the temperature if the setting is too high.	The temperature setting is easy to control manually.
	It didn't cost a fortune.	The thermostat is affordable to purchase.
Dislikes—current model	I'm too lazy to learn how to figure it out.	The thermostat requires little or no user instruction or learning.
	I sometimes forget to turn it off when we leave the house.	The thermostat saves energy when no one is home. (!)
	Sometimes the buttons don't register a push and I have to press it repeatedly.	The thermostat definitively registers any user inputs.
Suggested improvements	I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.
	I like to shift quickly between different options like energy saving or ultra comfort.	The thermostat responds immediately to differences in occupant preferences.

EXHIBIT 5-6 Customer data template filled in with sample customer statements and interpreted needs. (Note that this template represents a partial list from a single interview. A typical interview session may elicit more than 50 customer statements and interpreted needs.)

The final task in step 1 is to write thank-you notes to the customers involved in the process. Invariably, the team will need to solicit further customer information, so developing and maintaining a good rapport with a set of users is important.

Step 2: Interpret Raw Data in Terms of Customer Needs

Customer needs are expressed as written statements and are the result of interpreting the need underlying the raw data gathered from the customers. Each statement or observation (as listed in the second column of the data template) may be translated into any number of customer needs. Griffin and Hauser found that multiple analysts may translate the same interview notes into different needs, so it is useful to have more than one team member conducting the translation process. Later in this section, we provide five guidelines for writing needs statements. The first two guidelines are fundamental and are critical to effective translation; the remaining three guidelines ensure consistency of phrasing and style across all needs statements. Exhibit 5-7 provides examples to illustrate each guideline.

Guideline	Customer Statement	Needs Statement— Right	Needs Statement— Wrong
“What” not “How”	I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.	The thermostat is accompanied by a downloadable iPhone app.
Specificity	I have different heating and cooling systems.	The thermostat can control separate heating and cooling systems.	The thermostat is versatile.
Positive not negative	I get tired of standing in front of my thermostat to program it.	The thermostat can be programmed from a comfortable position.	The thermostat does not require me to stand in front of it for programming.
An attribute of the product	I have to manually override the program if I’m home when I shouldn’t be.	The thermostat automatically responds to an occupant’s presence.	An occupant’s presence triggers the thermostat to automatically change modes.
Avoid “must” and “should”	I’m worried about how secure my thermostat would be if it were accessible online.	The thermostat controls are secure from unauthorized access.	The thermostat must be secure from unauthorized access.

EXHIBIT 5-7 Examples illustrating the guidelines for writing needs statements.

- **Express the need in terms of what the product has to do, not in terms of how it might do it.** Customers often express their preferences by describing a solution concept or an implementation approach; however, the needs statement should be expressed in terms independent of a particular technological solution.
- **Express the need as specifically as the raw data.** Needs can be expressed at many different levels of detail. To avoid loss of information, express the need at the same level of detail as the raw data.
- **Use positive, not negative, phrasing.** Subsequent translation of a need into a product specification is easier if the need is expressed as a positive statement. This is not a rigid guideline, because sometimes positive phrasing is difficult and awkward. For example, one of the needs statements in Exhibit 5-6 is “The thermostat does not require replacing batteries.” This need is more naturally expressed in a negative form.
- **Express the need as an attribute of the product.** Wording needs as statements about the product ensures consistency and facilitates subsequent translation into product specifications. Not all needs can be cleanly expressed as attributes of the product, however, and in most of these cases the needs can be expressed as attributes of the user of the product (e.g., “An occupant’s presence triggers the thermostat to automatically change modes.”).
- **Avoid the words must and should.** The words *must* and *should* imply a level of importance for the need. Rather than casually assigning a binary importance rating (*must* versus *should*) to the needs at this point, we recommend deferring the assessment of the importance of each need until step 4.

The list of customer needs is the superset of all the needs elicited from all the interviewed customers in the target market. Some needs may not be technologically realizable.

The constraints of technical and economic feasibility are incorporated into the process of establishing product specifications in subsequent development steps. (See Chapter 6, Product Specifications.) In some cases, customers will have expressed conflicting needs. At this point in the process the team does not attempt to resolve such conflicts, but simply documents both needs. Deciding how to address conflicting needs is one of the challenges of the subsequent concept development activities.

Step 3: Organize the Needs into a Hierarchy

The result of steps 1 and 2 is generally a list of 50 to 300 *needs statements*. Such a large number of detailed needs is awkward to work with and difficult to summarize for use in subsequent development activities. The goal of step 3 is to organize these needs into a useful hierarchical list. The list will typically consist of a set of *primary needs*, each one of which will be further characterized by a set of *secondary needs*. In cases of very complex products, the secondary needs may be broken down into tertiary needs as well. The primary needs are the most general needs, while the secondary and tertiary needs express needs in more detail. Exhibit 5-8 shows the resulting hierarchical list of needs for the thermostat example. In this instance, there are 15 primary needs and 49 secondary needs.

The procedure for organizing the needs into a hierarchical list is intuitive, and many teams can successfully complete the task without detailed instructions. For completeness, we provide a step-by-step procedure here. This activity is best performed on a wall or a large table by a small group of team members.

1. **Print or write each needs statement on a separate card or self-stick note.** A print macro can be easily written to print the needs statements directly from the data template. A nice feature of this approach is that the need can be printed in a large font in the center of the card and then the original customer statement and other relevant information can be printed in a small font at the bottom of the card for easy reference. Four cards can be cut from a standard printed sheet.

2. **Eliminate redundant statements.** Those cards expressing redundant needs statements can be stapled together and treated as a single card. Be careful to consolidate only those statements that are identical in meaning.

3. **Group the cards according to the similarity of the needs they express.** At this point, the team should attempt to create groups of roughly three to seven cards that express similar needs. The logic by which groups are created deserves special attention. Novice development teams often create groups according to a technological perspective, clustering needs relating to, for example, materials, packaging, or power. Or they create groups according to assumed components such as enclosure, display, dial, and software. Both of these approaches are dangerous. Recall that the goal of the process is to create a description of the needs of the customer. For this reason, the groupings should be consistent with the way customers think about their needs and not with the way the development team thinks about the product. The groups should correspond to needs customers would view as similar. In fact, some practitioners use a process in which customers actually organize the needs statements.

4. **For each group, choose a label.** The label is itself a statement of need that generalizes all of the needs in the group. It can be selected from one of the needs in the group, or the team can write a new needs statement.

**** The thermostat is easy to install.**

- *** The thermostat works with my existing heating and/or cooling system.
- ** The thermostat installation is an easy do-it-yourself project for a novice.
- ** The thermostat can control separate heating and cooling systems.
- * The thermostat can be installed without special tools.
The thermostat is easily purchased.

*** The thermostat lasts a long time.**

- The thermostat is safe to bump into.
- The thermostat resists dirt and dust.
- ! The thermostat exterior surfaces do not fade or discolor over time.
- The thermostat is recyclable at end of life.

***** The thermostat is easy to use.**

- ** The thermostat user interaction is easy to understand.
- * The thermostat is easy to learn to use.
- * The thermostat does not place significant demands on user memory.
- ! The thermostat can be programmed from a comfortable position.
The thermostat can be controlled remotely without requiring a special device.
- ! The thermostat works pretty well right out of the box with no set up.
The thermostat's behavior is easy to change.
The thermostat is easy to control manually.
The thermostat display is easy to read from a distance.
The thermostat display can be read clearly in all conditions.
The thermostat's controls accommodate users with limited dexterity.
The thermostat accommodates different conventions for temperature scales.
The thermostat accommodates different preferences for representing time and date.

**** The thermostat controls are precise.**

- ** The thermostat maintains temperature accurately.
The thermostat minimizes unintended variability in temperature.
The thermostat allows temperatures to be specified precisely.

***** The thermostat is smart.**

- *** The thermostat can adjust temperature during the day according to user preferences.
- ** The thermostat can be programmed to a precise schedule.
- ! The thermostat automatically responds to occupancy.
- ! The thermostat prevents pipes from freezing in cold months.
The thermostat alerts the user when a problem arises.
The thermostat does not require users to set time or date.
The thermostat adjusts automatically to the seasons.

*** The thermostat is personal.**

- * The thermostat accommodates different user preferences for comfort.
The thermostat accommodates different user preferences for energy efficiency.
The thermostat controls are secure from unauthorized access.
The thermostat provides useful information.

***** The thermostat is a good investment.**

- ** The thermostat is affordable to purchase.
- *** The thermostat saves energy.
- * The thermostat tracks cost savings.

**** The thermostat is reliable.**

- The thermostat does not require replacing batteries.
- The thermostat works normally when electric power is suspended.

EXHIBIT 5-8 Hierarchical list of primary and secondary customer needs for the thermostat. Importance ratings for the secondary needs are indicated by the number of *'s, with *** denoting critically important needs. Latent needs are denoted by !.

5. Consider creating supergroups consisting of two to five groups. If there are fewer than 20 groups, then a two-level hierarchy is probably sufficient to organize the data. In this case, the group labels are primary needs and the group members are secondary needs; however, if there are more than 20 groups, the team may consider creating supergroups, and therefore a third level in the hierarchy. The process of creating supergroups is

identical to the process of creating groups. As with the previous step, cluster groups according to similarity of the need they express and then create or select a supergroup label. These supergroup labels become the primary needs, the group labels become the secondary needs, and the members of the groups become tertiary needs.

6. Review and edit the organized needs statements. There is no single correct arrangement of needs in a hierarchy. At this point, the team may wish to consider alternative groupings or labels and may engage another team to suggest alternative arrangements.

The process is more complicated when the team attempts to reflect the needs of two or more distinct market segments. There are at least two approaches that can be taken to address this challenge. First, the team can label each need with the segment (and possibly the name) of the customer from whom the need was elicited. This way, differences in needs across segments can be observed directly. One practical visual technique for this labeling is to use different colors of paper for the cards on which the needs statements are written, with each color corresponding to a different market segment. The other approach to multiple market segments is to perform the clustering process separately for each market segment. Using this approach, the team can observe differences both in the needs themselves and in the ways in which these needs are best organized. We recommend that the team adopt this parallel, independent approach when the segments are very different in their needs and when there is some doubt about the ability of the team to address the different segments with the same product.

Step 4: Establish the Relative Importance of the Needs

The hierarchical list alone does not provide any information on the relative importance that customers place on different needs. Yet the development team will have to make trade-offs and allocate resources in designing the product. A sense of the relative importance of the various needs is essential to making these trade-offs correctly. Step 4 in the needs process establishes the relative importance of the customer needs identified in steps 1 through 3. The outcome of this step is a numerical importance weighting for a subset of the needs. There are two basic approaches to the task: (1) relying on the consensus of the team members based on their experience with customers, or (2) basing the importance assessment on further customer surveys. The obvious trade-off between the two approaches is cost and speed versus accuracy: the team can make an educated assessment of the relative importance of the needs in one meeting, while a customer survey generally takes a week or more. In most cases we believe the customer survey is important and worth the time required to complete it. Other development tasks, such as concept generation and analysis of competitive products, can begin before the relative importance surveys are complete.

At this point the team should have developed a rapport with a group of customers. These same customers can be surveyed to rate the relative importance of the needs that have been identified. The survey can be done in person, by telephone, via the Internet, or by mail. Few customers will respond to a survey asking them to evaluate the importance of 100 needs, so typically the team will work with only a subset of the needs. A practical limit on how many needs can be addressed in a customer survey is about 50. This limit is not too severe, however, because many of the needs are either obviously important (e.g., The thermostat maintains temperature accurately) or are easy to implement (e.g., The thermostat allows temperatures to be specified precisely). The team can therefore limit the scope of the survey by querying customers only about needs that are likely to give rise to difficult technical

Thermostat Survey

For each of the following thermostat features, please indicate on a scale of 1 to 5 how important the feature is to you. Please use the following scale:

1. Feature is undesirable. I would not consider a product with this feature.
2. Feature is not important, but I would not mind having it.
3. Feature would be nice to have, but is not necessary.
4. Feature is highly desirable, but I would consider a product without it.
5. Feature is critical. I would not consider a product without this feature.

Also indicate by checking the box to the right if you feel that the feature is unique, exciting, and/or unexpected.

Importance of feature
on scale of 1 to 5

Check box if feature is unique,
exciting, and/or unexpected.

- | | |
|---|--------------------------|
| _____ The thermostat does not require the user to set time or date. | ☐ |
| _____ The thermostat does not require replacing batteries. | ☐ |
| _____ The thermostat adjusts automatically to the seasons. | ☐ |
| _____ The thermostat accommodates different preferences for representing date and time. | ☐ |

And so forth.

EXHIBIT 5-9 Example importance survey (partial).

trade-offs or costly features in the product design. Such needs would include, for instance, the need to avoid replacing batteries, or the need to not have to enter time or date. Alternatively the team could develop a set of surveys to ask a variety of customers each about different subsets of the needs list. There are many survey designs for establishing the relative importance of customer needs. One good design is illustrated by the portion of the thermostat survey shown in Exhibit 5-9. In addition to asking for importance ratings, this survey asks the respondent to explicitly identify the needs that are unique or unexpected. This information can be used to help the team identify latent needs.

The survey responses for each needs statement can be characterized in a variety of ways: by the mean, by the standard deviation, or by the number of responses in each category. The responses can then be used to assign an importance weighting to the needs statements. The same scale of 1 to 5 can be used to summarize the importance data. The needs in Exhibit 5-8 are rated according to the survey data, with the importance ratings denoted by the number of *'s next to each needs statement and the latent needs denoted by !. Note that no critical needs are also latent needs. This is because if a need were critical, customers would not be surprised or excited by it; they would expect it to be met.

Step 5: Reflect on the Results and the Process

The final step in the method is to reflect on the results and the process. While the process of identifying customer needs can be usefully structured, it is not an exact science. The team must challenge its results to verify that they are consistent with the knowledge and intuition the team has developed through many hours of interaction with customers. Some questions to ask include:

- Have we interacted with all of the important types of customers in our target market?
- Are we able to see beyond needs related only to existing products to capture the latent needs of our target customers?

- Are there areas of inquiry we should pursue in follow-up interviews or surveys?
- Which of the customers we spoke to would be good participants in our ongoing development efforts?
- What do we know now that we didn't know when we started? Are we surprised by any of the needs?
- Did we involve everyone within our own organization who needs to deeply understand customer needs?
- How might we improve the process in future efforts?

A good way to summarize the results of the process is to list the most important needs and the latent needs. For instance, the most important needs identified are:

- The thermostat is easy to use.
- The thermostat can adjust temperature during the day according to user preferences.
- The thermostat works with my existing heating and/or cooling system.
- The thermostat reduces energy consumption.

The latent needs are:

- The thermostat can be programmed from a comfortable position.
- The thermostat works pretty well right out of the box with no set up.
- The thermostat automatically responds to occupancy.
- The thermostat exterior surfaces do not fade or discolor over time.
- The thermostat prevents pipes from freezing in cold months.

The important needs are an excellent reminder of the critical few needs that must be addressed for a great product, and the latent needs provide insights that can drive the creative process of generating product concepts.

Summary

Identifying customer needs is an integral part of the concept development phase of the product development process. The resulting customer needs are used to guide the team in establishing product specifications, generating product concepts, and selecting a product concept for further development.

- The process of identifying customer needs includes five steps:
 1. Gather raw data from customers.
 2. Interpret the raw data in terms of customer needs.
 3. Organize the needs into a hierarchy.
 4. Establish the relative importance of the needs.
 5. Reflect on the results and the process.
- Creating a high-quality information channel from customers to the product developers ensures that those who directly control the details of the product, including the product designers, fully understand the needs of the customer.

- Lead users are a good source of customer needs because they experience new needs months or years ahead of most customers and because they stand to benefit substantially from new product innovations. Furthermore, they are frequently able to articulate their needs more clearly than typical customers. Extreme users have special needs that may reflect latent needs among mainstream users.
- Latent needs may be even more important than explicit needs in determining customer satisfaction. Latent needs are those that many customers recognize as important in a final product but do not or cannot articulate in advance.
- Customer needs should be expressed in terms of what the product has to do, not in terms of how the product might be implemented. Adherence to this principle leaves the development team with maximum flexibility to generate and select product concepts.
- The key benefits of the method are ensuring that the product is focused on customer needs and that no critical customer need is forgotten; developing a clear understanding among members of the development team of the needs of the customers in the target market; developing a fact base to be used in generating concepts, selecting a product concept, and establishing product specifications; and creating an archival record of the needs phase of the development process.

References and Bibliography

Many current resources are available on the Internet via

www.ulrich-eppinger.net

Concept engineering is a method developed by Burchill at MIT in collaboration with the Center for Quality of Management. This chapter benefits from our observations of the development and application of concept engineering. For a complete and detailed description of concept engineering, see:

Burchill, Gary, et al., *Concept Engineering*, Center for Quality of Management, Cambridge, MA, Document No. ML0080, 1997.

The research by Griffin and Hauser validates different methods for extracting needs from interview data. Their study of the fraction of needs identified as a function of the number of customers interviewed is particularly interesting.

Griffin, Abbie, and John R. Hauser, "The Voice of the Customer," *Marketing Science*, Vol. 12, No. 1, Winter 1993, pp. 1–27.

Kinney and Taylor thoroughly discuss data collection methods and survey design.

Kinney, Thomas C., and James R. Taylor, *Marketing Research: An Applied Approach*, fifth edition, McGraw-Hill, New York, 1995.

Norman has written extensively on user needs, especially as related to the cognitive challenges of using products.

Norman, Donald A., *The Design of Everyday Things*, Doubleday, New York, 1990.

Payne's book is a detailed and interesting discussion of how to pose questions in surveys.

Payne, Stanley L., *The Art of Asking Questions*, Princeton University Press, Princeton, NJ, 1980.

Total quality management (TQM) provides a valuable perspective on how identifying customer needs fits into an overall effort to improve the quality of goods and services.

Shiba, Shoji, Alan Graham, and David Walden, *A New American TQM: Four Practical Revolutions in Management*, Productivity Press, Cambridge, MA, and The Center for Quality of Management, Cambridge, MA, 1993.

Urban and Hauser provide a thorough discussion of how to create hierarchies of needs (along with many other topics).

Urban, Glen L., and John R. Hauser, *Design and Marketing of New Products*, second edition, Prentice Hall, Englewood Cliffs, NJ, 1993.

Von Hippel describes many years of research on the role of lead users in innovation. He provides useful guidelines for identifying lead users.

von Hippel, Eric, *The Sources of Innovation*, Oxford University Press, New York, 1988.

Exercises

1. Translate the following customer statements about a student book bag into proper needs statements:
 - a. “See how the leather on the bottom of the bag is all scratched; it’s ugly.”
 - b. “When I’m standing in line at the cashier trying to find my checkbook while balancing my bag on my knee, I feel like a stork.”
 - c. “This bag is my life; if I lose it I’m in big trouble.”
 - d. “There’s nothing worse than a banana that’s been squished by the edge of a textbook.”
 - e. “I never use both straps on my knapsack; I just sling it over one shoulder.”
2. Observe someone performing an everyday task. (Ideally, you should choose a task for which you can observe different users performing the task repeatedly.) Identify frustrations and difficulties encountered by these people. Identify the latent customer needs.
3. Choose a product that continually annoys you. Identify the needs the developers of this product missed. Why do you think these needs were not met? Do you think the developers deliberately ignored these needs?

Thought Questions

1. One of the reasons the method is effective is that it involves the entire development team. Unfortunately, the method can become unwieldy with a team of more than 10 people. How might you modify the method to maximize involvement yet maintain a focused and decisive effort given a large development team?
2. Can the process of identifying customer needs lead to the creation of innovative product concepts? In what ways? Could a structured process of identifying customer needs lead to a fundamentally new product concept like the Post-it Note?